

Lab 4: Web API CRUD Operation (PUT)

Objective

To implement the HTTP PUT action method in an ASP.NET Core Web API to update employee information using the FromBody attribute, validate employee IDs, return appropriate BadRequest responses for invalid IDs, and test the API using Swagger and Postman.

Code

```
Program.cs Employee.cs 3 Department.cs 1 Skill.cs 1 EmployeeController.cs >
Controllers > EmployeeController.cs > {} FirstWebAPI.Controllers > EmployeeController > UpdateEmployee(Em
6 namespace FirstWebAPI.Controllers;
11 public class EmployeeController : ControllerBase
95 public IActionResult AddEmployee([FromBody] Employee employee)
97     employees.Add(employee);
98
99     return Ok(employee);
100 }
101
102 [HttpPut]
0 references
103 public ActionResult<Employee> UpdateEmployee([FromBody] Employee employee)
104 {
105     // Check if the employee ID is valid
106     if (employee.Id <= 0)
107     {
108         return BadRequest("Invalid employee id");
109     }
110
111     // Find the employee in the hardcoded list
112     var existingEmployee = employees.FirstOrDefault(e => e.Id == employee.Id);
113
114     if (existingEmployee == null)
115     {
116         return BadRequest("Invalid employee id");
117     }
118
119     // Update the employee details
120     existingEmployee.Name = employee.Name;
121     existingEmployee.Salary = employee.Salary;
122     existingEmployee.Permanent = employee.Permanent;
123     existingEmployee.Department = employee.Department;
124     existingEmployee.Skills = employee.Skills;
125     existingEmployee.DateOfBirth = employee.DateOfBirth;
126
127     // Return the updated employee
128     return Ok(existingEmployee);
129 }
```

Output

```
type.  
C:\Users\laksh\FirstWebAPI\Filters\CustomAuthFilter.cs(18,18): warning CS8602: Dereferencing a null value.  
Build succeeded with 7 warning(s) in 1.6s  
C:\Users\laksh\FirstWebAPI> dotnet run  
Using launch settings from C:\Users\laksh\FirstWebAPI\Properties\launchSettings.json...  
Building...  
Info: Microsoft.Hosting.Lifetime[14]  
Now listening on: http://localhost:5074  
Info: Microsoft.Hosting.Lifetime[0]  
Application started. Press Ctrl+C to shut down.  
Info: Microsoft.Hosting.Lifetime[0]  
Hosting environment: Development  
Info: Microsoft.Hosting.Lifetime[0]  
Content root path: C:\Users\laksh\FirstWebAPI
```

 **Swagger**
Supported by SMARTBEAR

FirstWebAPI

1.0 OAS 3.0

<http://localhost:5074/swagger/v1/swagger.json>

Employee

GET

/api/Employee

POST

/api/Employee

PUT

/api/Employee

Code	Description
200	OK Media type text/plain Controls Accept header. Example Value Schema

